BAO Sound Horizon and Baryon Loading Resolution

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Abstract. We resolve the BAO sound horizon tension by correcting the baryon loading formulation in the early universe. We show that when the neutrino density is excluded from the baryon loading denominator, standard GR with SDGFT density parameters yields a sound horizon of rd 147.07 Mpc, matching Planck data within 0.1 sigma.

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1 Introduction

The BAO sound horizon is a key anchor for late-time cosmology. We show that previous estimates of tension were due to a baryon loading bug.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$r_d = \int_z^\infty \frac{c_s(z')}{H(z')} dz' \approx 147.07 \text{ Mpc} \tag{1}$$

$$c_s(z) = \frac{c}{\sqrt{3\left(1 + \frac{3\Omega_b}{4\Omega_\gamma(1+z)}\right)}} \tag{2}$$

3 Baryon Loading Bug Correction

By separating the photon and neutrino densities, we show that the sound horizon converges to 147.07 Mpc under the SDGFT background.

4 Conclusion

The BAO tension is resolved, demonstrating the absolute consistency of standard GR with the topological density parameters of SDGFT.

References

[1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).

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[3] D. A. Besemer, `sdgft` Python package, https://github.com/david-besemer/sdgft (2026).